

Differential Equations

Final Exam

(MT1006)

Date: May 29th 2024

Course Instructor(s)

Dr. Tayyaba Naz

Total Time (Hrs):

3

Total Marks:

90

Total Questions:

4

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

CLO #2: Determine the solutions of 1st order DEs and higher order DEs

- Q1 (a) Find the general solution of the given linear differential equation. Give the largest interval I over which the general solution is defined. Determine whether there are any transient terms in the general solution. [10 marks]

$$x^2 y' + x(x+2)y = e^x.$$

- (b) Solve given differential equation by using undetermined coefficients approach [10 marks]

$$y = e^{mx}$$

$$y'' + 2y' = 2x + 5 - e^{-2x}.$$

$$y = c_1 e^{-2x} + c_2 + 5x - \frac{1}{2} e^{-2x}$$

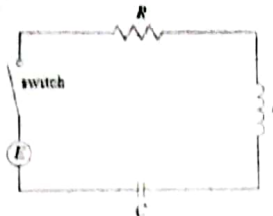
- (c) Identify the given differential equation and find the general solution of given differential equation. [10 marks]

$$y = x^{4.7}$$

$$x^2 y'' + xy' - y = \ln x.$$

CLO #3: Formulate DEs with applications

- Q2: A 100-volt electromotive force is applied to an RC-series circuit in which the resistance is 200 ohms and the capacitance is 10^{-4} farad. Find the charge $q(t)$ on the capacitor if $q(0) = 0$. Also find the current $i(t)$. The linear differential equation for the current $i(t)$ is given as $L \frac{di}{dt} + Ri = E(t)$.



[10 marks]

CLO #4: Solve problems using Laplace Transform

- Q3: (a) Use definition method to evaluate $L\{f(t)\}$, where [10 marks]

$$f(t) = \begin{cases} 2t + 1, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}$$

- b) Use Laplace transform to solve the given initial value problem. [10 marks]

7.2

$$y'' - 6y' + 13y = 0, \quad y(0) = 0, \quad y'(0) = -3.$$

Spring 2024

Department of Electrical Engineering

Page 1 of 2

$$\begin{aligned} &= \int_0^{\infty} e^{-st} \cdot f(t) dt \\ &= \int_0^1 e^{-st} (2t+1) dt \end{aligned}$$

National University of Computer and Emerging Sciences
Lahore Campus

(c) Evaluate the following expressions by using appropriate formula.

[15 marks]

7.3

i. $\mathcal{L}\{e^t \sin 3t\}$

ii. $\mathcal{L}\{t \mu(t-2)\}$

iii. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$

CLO #5: Use the concept of series to appraise the solutions of DEs.

Q4: (a) Find two power series solutions of given differential equation about the ordinary point $x = 0$.

6.1

$$(x^2 + 2)y'' + 3xy' - y = 0.$$

[10 marks]

6.2

(b) Determine the singular points of the given differential equation. Classify each singular point as regular or irregular.

$$(x^2 - 9)^2 y'' + (x+3)y' + 2y = 0.$$

[5 marks]

Formula Sheet:

$$(a) \mathcal{L}\{1\} = \frac{1}{s}$$

$$(b) \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 1, 2, 3, \dots \quad (c) \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$(d) \mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2} \quad (e) \mathcal{L}\{\cos kt\} = \frac{s}{s^2 + k^2}$$

$$(f) \mathcal{L}\{\sinh kt\} = \frac{k}{s^2 - k^2} \quad (g) \mathcal{L}\{\cosh kt\} = \frac{s}{s^2 - k^2}$$

$$(a) 1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$$

$$(b) t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\}, \quad n = 1, 2, 3, \dots \quad (c) e^{at} = \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\}$$

$$(d) \sin kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\} \quad (e) \cos kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + k^2}\right\}$$

$$(f) \sinh kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2 - k^2}\right\} \quad (g) \cosh kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2 - k^2}\right\}$$

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0),$$

where $F(s) = \mathcal{L}\{f(t)\}$.